

# Fine-Tuning Large Language Models for Structured Dashboard Design Recommendations Based on User Goals and Data Context

Ramin Saeedi Valashani  
M.Sc. Computer Science  
Institute of Computer Science  
Hochschule Ruhr West, Bottrop, Germany  
ramin.saeedivalashani@stud.hs-ruhrwest.de

**Abstract**—Dashboards support data-driven decisions, yet designing effective dashboards requires expertise in data visualization and HCI. Generic large language models (LLMs) can provide advice, but their outputs are often vague, inconsistent, and not reliably grounded in best practices. This proposal investigates whether parameter-efficient fine-tuning (PEFT) of an open-source LLM improves the quality of structured, context-aware dashboard design recommendations given a short “dashboard brief” describing target users, goals, KPIs, and data characteristics. In addition to prompt-only baselines, the study will include retrieval-augmented generation (RAG) to ground recommendations in external guideline knowledge. We compare (i) prompt-only, (ii) RAG, (iii) fine-tuning, and (iv) fine-tuning+RAG using chart recommendation accuracy (Top- $k$  where reference labels exist), schema compliance, and human rubric ratings with inter-rater reliability and statistical testing.

**Index Terms**—dashboards, data visualization, large language models, fine-tuning, PEFT, LoRA, QLoRA, RAG

## I. MOTIVATION

Dashboards communicate KPIs and operational signals at a glance. Poor design can cause cognitive overload, hide important trends, and lead to misinterpretation. Practitioners must choose chart types that match analytical tasks (trend, comparison, composition, distribution), establish a clear information hierarchy, and keep interfaces readable. While visualization and HCI research provide design principles [1], [2], [3], applying them consistently remains difficult in practice.

LLMs are attractive as “design assistants” because they can transform a textual requirement into guidance. However, out-of-the-box LLM recommendations are frequently generic, inconsistent across similar inputs, or weakly justified. This motivates domain adaptation via fine-tuning and grounding via retrieval.

## II. PROBLEM STATEMENT

There is currently no lightweight assistant that reliably converts a short dashboard brief into a structured, actionable design plan. Prompt-only approaches commonly fail in three ways: (i) missing structure (e.g., omitted task-to-chart mapping), (ii) low specificity, and (iii) weak grounding in accepted heuristics.

This thesis focuses on generating a text-based design brief (not a pixel-perfect UI). The output will follow a fixed schema (JSON or a sectioned template) that includes: (1) context summary, (2) KPI/task  $\rightarrow$  chart mapping, (3) layout and information hierarchy, (4) labels/scales/colors (incl. accessibility notes), (5) interaction suggestions (filters/drill-down), and (6) short design rationales.

## III. RESEARCH QUESTIONS

RQ1 (Accuracy): How accurately can the system map dashboard tasks/KPIs to suitable chart types (Top-1/Top-3 where reference labels exist) across prompt-only, RAG, fine-tuning, and fine-tuning+RAG?

RQ2 (Actionability & Structure): Do RAG and/or fine-tuning improve actionability and structure, measured via schema compliance and human rubric ratings?

RQ3 (Rationale & Robustness): Do RAG and/or fine-tuning improve heuristic-aligned rationales and robustness under paraphrased or partially specified briefs?

## IV. METHODOLOGY

### A. Rubric and Output Schema

A rubric with 8–10 dimensions will be compiled from authoritative visualization and HCI sources. Example dimensions include task–chart fit, hierarchy, readability/labeling, scale correctness, color/accessibility, information density, interaction relevance, and rationale quality. The model output will be constrained by a fixed schema to enable automatic checking and consistent evaluation.

### B. Data and Fine-Tuning (Model Scale)

Only publicly available datasets will be used; no new human-labeled dataset will be created. Existing data will be transformed into prompt–response pairs that match the target task (dashboard brief  $\rightarrow$  structured recommendation). To address model-capacity concerns, experiments will include at least one larger base model (e.g.,  $\geq 13$ B class) in addition to a medium model, depending on compute availability. PEFT methods such as LoRA/QLoRA enable efficient adaptation of large models [4], [5].

### C. Retrieval-Augmented Generation (RAG)

RAG combines a parametric LLM with non-parametric knowledge via retrieval [6], [7]. This project will build a retrieval corpus from publicly available dashboard/visualization guidelines and structured references (e.g., chart-selection guidance, labeling, accessibility). At inference time, the system retrieves the top- $k$  most relevant guideline passages for a given dashboard brief and conditions generation on them (RAG). RAG is widely used to improve grounding and reduce hallucinations in generation tasks [6], [7], [8], [9], [10].

### D. Baselines and Systems

We evaluate four systems: (A) prompt-only base model with schema, (B) prompt-only + RAG grounding, (C) PEFT fine-tuned model, and (D) PEFT fine-tuned model + RAG. This isolates the individual and combined effects of grounding and fine-tuning [6], [7], [11].

## V. EVALUATION PLAN

### A. Automatic Metrics

Primary automatic metrics are: (i) chart-type recommendation performance (Top-1/Top-3 accuracy when reference labels exist, plus macro-F1 over chart-type classes), (ii) schema compliance for structured outputs (valid JSON if used; required fields/sections present; completeness score as fraction of required fields), and (iii) robustness/consistency under perturbations (paraphrased briefs and missing-information variants), following behavioral testing practices for NLP models [12]. For RAG-based systems, we additionally report a grounding proxy (unsupported-claim rate) by checking whether rationale statements are supported by retrieved guideline passages, motivated by hallucination/faithfulness findings [10]. Automatic metrics are used as supportive evidence and interpreted in line with known limitations of NLG evaluation [11], [13].

### B. Human Evaluation and Sample Size

Human evaluation will follow best-practice guidance for NLP user studies and statistical evaluation [11], [14]. We plan a within-subject design with at least  $N = 60$ –100 dashboard briefs/items (target: 100, consistent with typical NLG evaluations [14]). To obtain sufficient statistical power, we will recruit 6–10 raters (mix of visualization/UX-aware students and/or practitioners) and use a balanced assignment so that each item is rated by at least 3 independent raters. Raters score outputs on the rubric (Likert 1–5). Inter-rater reliability will be reported using Krippendorff’s alpha [15]. To control workload and avoid fatigue effects, we will use a balanced incomplete block design so that each rater evaluates only a subset of items, while maintaining at least three ratings per item/system. We will run a small pilot (e.g., 10–15 items) to estimate rating variance and confirm the planned sample size for adequate statistical power (target power  $\approx 0.8$ ) [11].

### C. Statistical Significance

For Likert (ordinal) rubric ratings across the four systems, we will use the Friedman test for repeated measures [16] followed by post-hoc paired Wilcoxon signed-rank tests [17] with multiplicity correction (Holm or Bonferroni) [11], [18]. We will report effect sizes suitable for ordinal data (e.g., Cliff’s  $\delta$ ) [19] and 95% confidence intervals (bootstrap over items). For binary correctness outcomes (e.g., Top-1 correct / incorrect) across multiple systems, we will use Cochran’s  $Q$  test [20] and conduct post-hoc paired McNemar tests [21] with multiplicity correction.

## VI. EXPECTED CONTRIBUTIONS

The expected outcomes are: (1) a fine-tuned, grounded system that produces more specific and actionable dashboard design briefs, (2) a reusable rubric and evaluation protocol, and (3) empirical evidence of the added value of RAG, fine-tuning, and their combination for dashboard design recommendation quality.

## VII. TIMELINE (6 MONTHS)

- Month 1: Literature review; define rubric and output schema.
- Month 2: Dataset selection and preprocessing; build retrieval corpus and index; baseline prompts.
- Month 3: Fine-tuning runs; RAG integration; hyperparameter tuning.
- Month 4: Automatic and human evaluation; reliability analysis.
- Month 5: Error analysis and robustness tests; finalize prototype/demo.
- Month 6: Writing and defense preparation.

## VIII. RISKS

Key risks include dataset-task mismatch, limited compute, and subjective human ratings. Mitigation includes strict output schemas, PEFT methods, careful retrieval corpus design, reviewer calibration, and reliability reporting.

## REFERENCES

- [1] S. Few, *Information Dashboard Design: The Effective Visual Communication of Data*. O’Reilly Media, 2006.
- [2] T. Munzner, *Visualization Analysis and Design*. A K Peters/CRC Press, 2014. DOI: [10.1201/b17511](https://doi.org/10.1201/b17511).
- [3] W. S. Cleveland and R. McGill, “Graphical perception: Theory, experimentation, and application to the development of graphical methods,” *Journal of the American Statistical Association*, vol. 79, no. 387, pp. 531–554, 1984. DOI: [10.1080/01621459.1984.10478080](https://doi.org/10.1080/01621459.1984.10478080).
- [4] E. J. Hu et al., “Lora: Low-rank adaptation of large language models,” *arXiv preprint arXiv:2106.09685*, 2021. [Online]. Available: <https://arxiv.org/abs/2106.09685>.
- [5] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “Qlora: Efficient finetuning of quantized llms,” *arXiv preprint arXiv:2305.14314*, 2023. [Online]. Available: <https://arxiv.org/abs/2305.14314>.

- [6] P. Zhao et al., "Retrieval-augmented generation for ai-generated content: A survey," *Data Science and Engineering*, 2026. DOI: [10.1007/s41019-025-00335-5](https://doi.org/10.1007/s41019-025-00335-5). [Online]. Available: <https://link.springer.com/article/10.1007/s41019-025-00335-5>.
- [7] C. Sharma, "Retrieval-augmented generation: A comprehensive survey of architectures, enhancements, and robustness frontiers," *arXiv preprint arXiv:2506.00054*, 2025. [Online]. Available: <https://arxiv.org/abs/2506.00054>.
- [8] A. Singh, A. Ehtesham, S. Kumar, and T. T. Khoei, "Agentic retrieval-augmented generation: A survey on agentic rag," *arXiv preprint arXiv:2501.09136*, 2025. [Online]. Available: <https://arxiv.org/abs/2501.09136>.
- [9] M. Du et al., "A-rag: Scaling agentic retrieval-augmented generation via hierarchical retrieval interfaces," *arXiv preprint arXiv:2602.03442*, 2026. [Online]. Available: <https://arxiv.org/abs/2602.03442>.
- [10] L. Huang et al., "A survey on hallucination in large language models," *ACM Computing Surveys*, 2025. DOI: [10.1145/3703155](https://doi.org/10.1145/3703155). [Online]. Available: <https://dl.acm.org/doi/10.1145/3703155>.
- [11] *How to do human evaluation: Best practices for user studies in nlp*, OpenReview, 2023. [Online]. Available: <https://openreview.net/pdf?id=h08TpsSm0bl>.
- [12] M. T. Ribeiro, T. Wu, C. Guestrin, and S. Singh, "Beyond accuracy: Behavioral testing of NLP models with CheckList," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Online: Association for Computational Linguistics, Jul. 2020, pp. 4902–4912. DOI: [10.18653/v1/2020.acl-main.442](https://doi.org/10.18653/v1/2020.acl-main.442). [Online]. Available: <https://aclanthology.org/2020.acl-main.442/>.
- [13] S. Gehrmann, E. Clark, and T. Sellam, "Repairing the cracked foundation: A survey of obstacles in evaluation practices for generated text," *arXiv preprint arXiv:2202.06935*, 2022. DOI: [10.48550/arXiv.2202.06935](https://doi.org/10.48550/arXiv.2202.06935). [Online]. Available: <https://arxiv.org/abs/2202.06935>.
- [14] C. van der Lee, A. Gatt, E. van Miltenburg, S. Wubben, and E. Krahmer, "Best practices for the human evaluation of automatically generated text," in *Proceedings of the 12th International Conference on Natural Language Generation (INLG)*, 2019. [Online]. Available: <https://aclanthology.org/W19-8643/>.
- [15] K. Krippendorff, "Computing krippendorff's alpha-reliability," University of Pennsylvania, Tech. Rep., 2011. [Online]. Available: <https://www.asc.upenn.edu/sites/default/files/2021-03/Computing%20Krippendorff%27s%20Alpha-Reliability.pdf>.
- [16] M. Friedman, "The use of ranks to avoid the assumption of normality implicit in the analysis of variance," *Journal of the American Statistical Association*, vol. 32, no. 200, pp. 675–701, 1937. DOI: [10.1080/01621459.1937.10503522](https://doi.org/10.1080/01621459.1937.10503522). [Online]. Available: <https://doi.org/10.1080/01621459.1937.10503522>.
- [17] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945. DOI: [10.2307/3001968](https://doi.org/10.2307/3001968). [Online]. Available: <https://doi.org/10.2307/3001968>.
- [18] H.-Y. Kim, "Statistical notes for clinical researchers: Nonparametric statistical methods: 2. nonparametric methods for comparing three or more groups and repeated measures," *Restorative Dentistry & Endodontics*, 2014. DOI: [10.5395/rde.2014.39.4.329](https://doi.org/10.5395/rde.2014.39.4.329). [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4223105/>.
- [19] K. Meissel and E. S. Yao, "Using cliff's delta as a non-parametric effect size measure: An accessible web app and r tutorial," *Practical Assessment, Research, and Evaluation*, vol. 29, no. 1, p. 2, 2024. DOI: [10.7275/pare.1977](https://doi.org/10.7275/pare.1977). [Online]. Available: <https://doi.org/10.7275/pare.1977>.
- [20] W. G. Cochran, "The comparison of percentages in matched samples," *Biometrika*, vol. 37, no. 3–4, pp. 256–266, 1950. DOI: [10.1093/biomet/37.3-4.256](https://doi.org/10.1093/biomet/37.3-4.256). [Online]. Available: <https://doi.org/10.1093/biomet/37.3-4.256>.
- [21] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947. DOI: [10.1007/BF02295996](https://doi.org/10.1007/BF02295996). [Online]. Available: <https://doi.org/10.1007/BF02295996>.